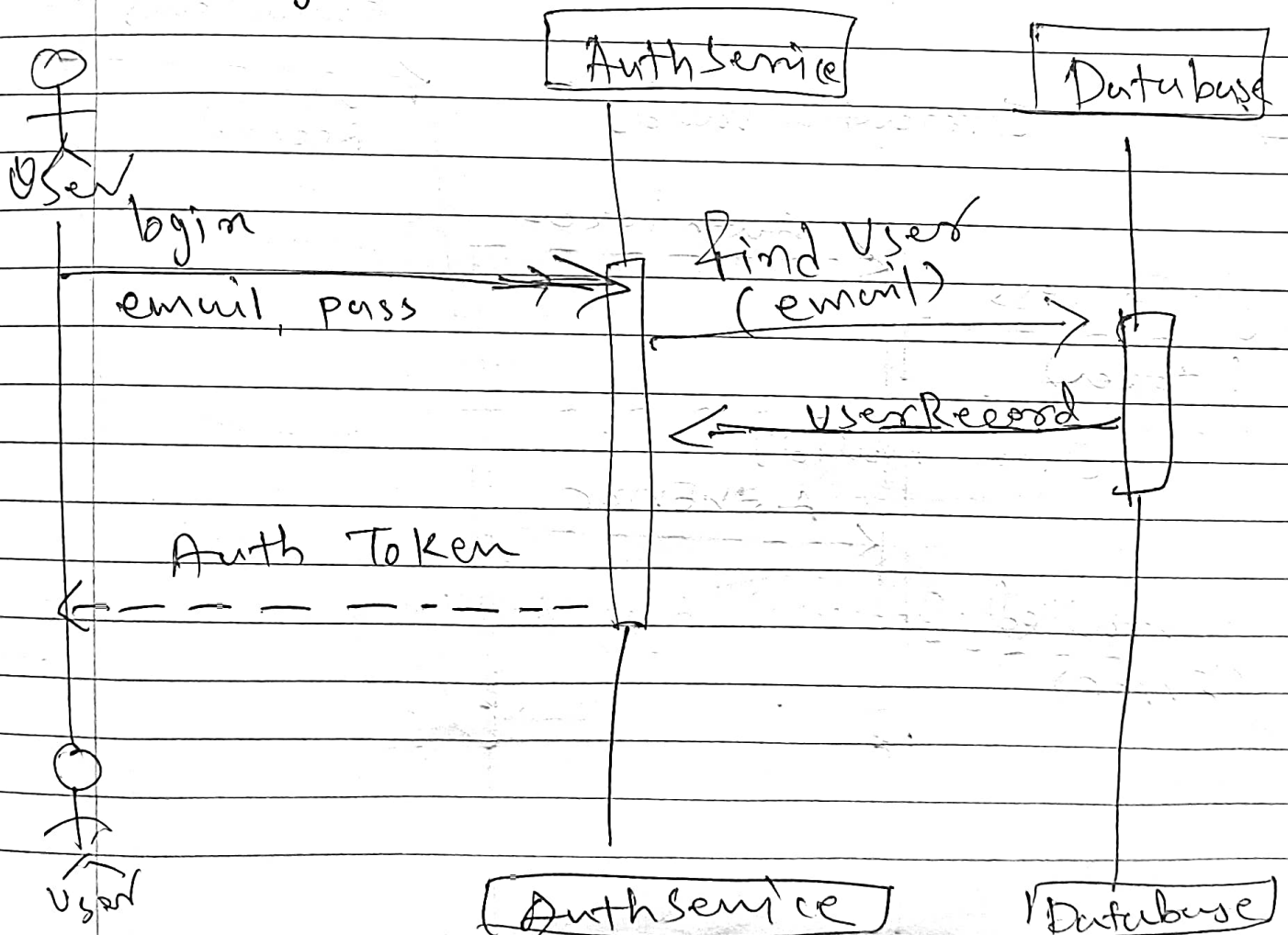


Sequence diagram.

- A sequence diagram is a type of UML diagram that shows how objects in a system interact with each other, step by step.
- It focuses on the order of messages exchanged between different components or actors to achieve a particular task or use case.
- These model:
 - who the participants (objects or actors) are
 - What messages are exchanged
 - In what order the messages occur
 - How long each participant is active.

Building blocks :

- Actors and participants.
- Lifelines
- Activation Bars
- Messages.



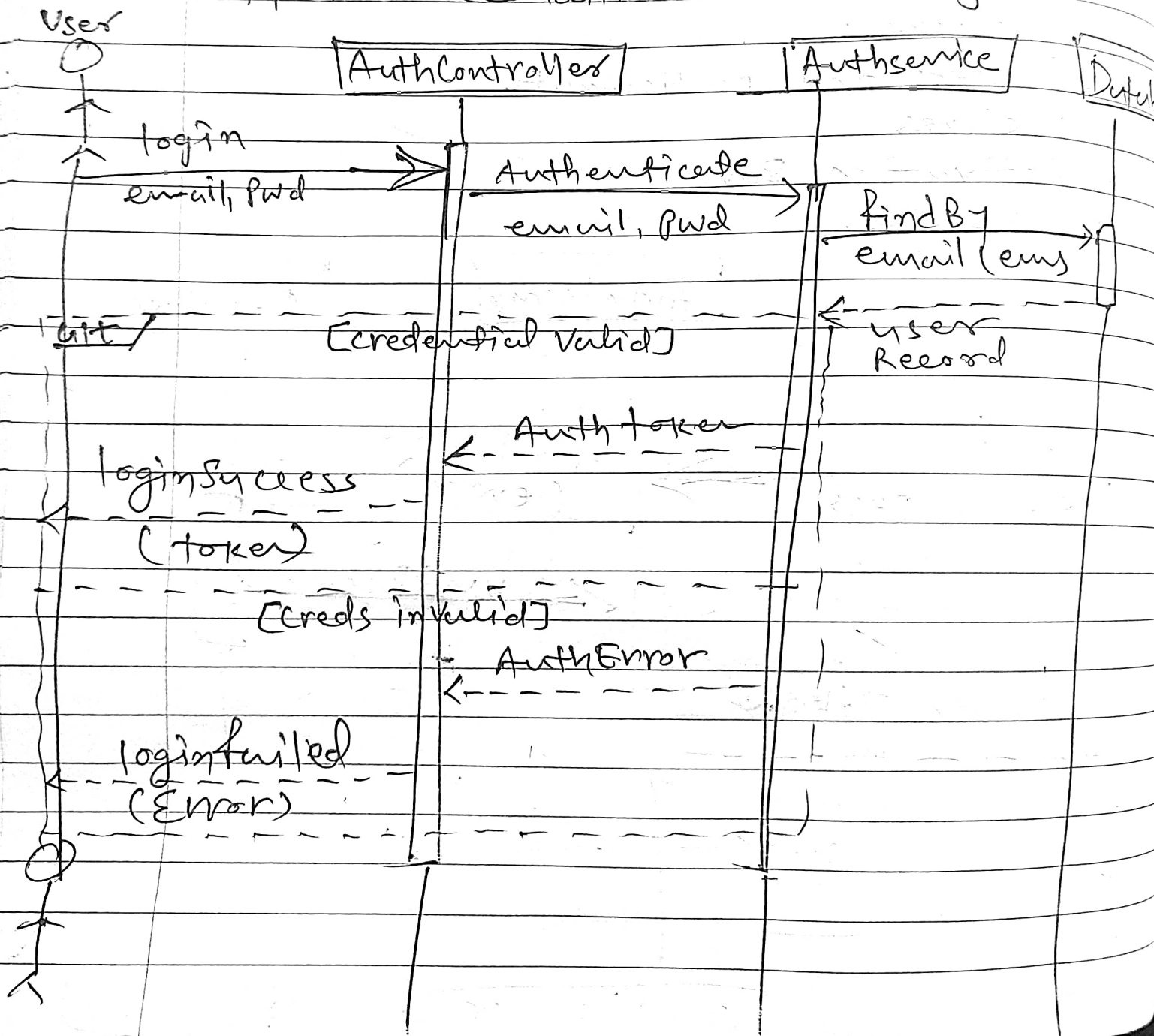
⇒ Types of Messages in seq diagrams

- Synchronous messages
- Async messages
- Return Message
- self-messages

- Create Messages
- Destroy Messages

- Combined Fragments

- alt/else (Conditional Branching)

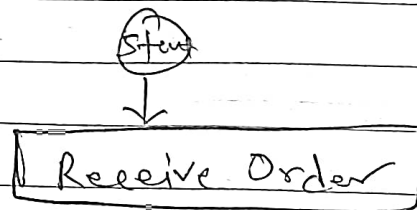


loop (Repetition)
opt (Optional)
par (parallel execution)

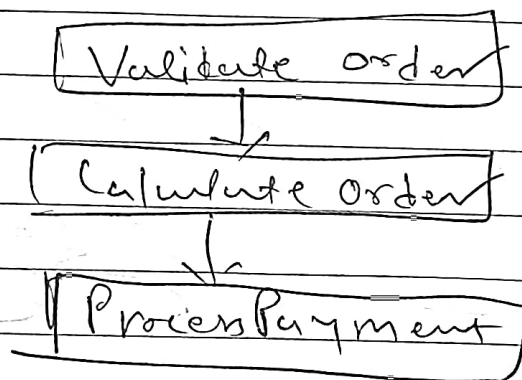
Activity Diagram

- An activity diagram is a UML diagram that visualizes a workflow. It shows
 - what activities/actions happen
 - In what order they execute
 - Where decisions are made
 - What can happen in parallel
 - Who is responsible (with swimlanes)

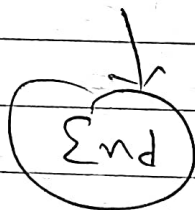
Initial Node:



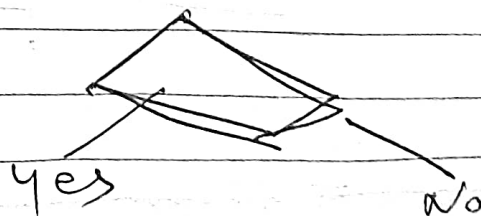
Activity / Action Node:



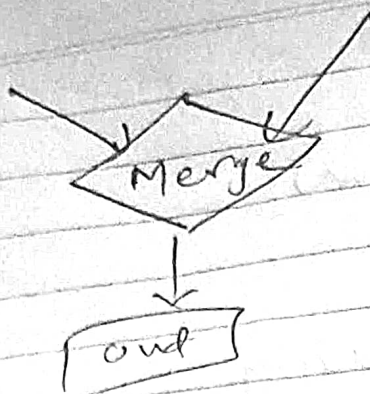
Final Node:



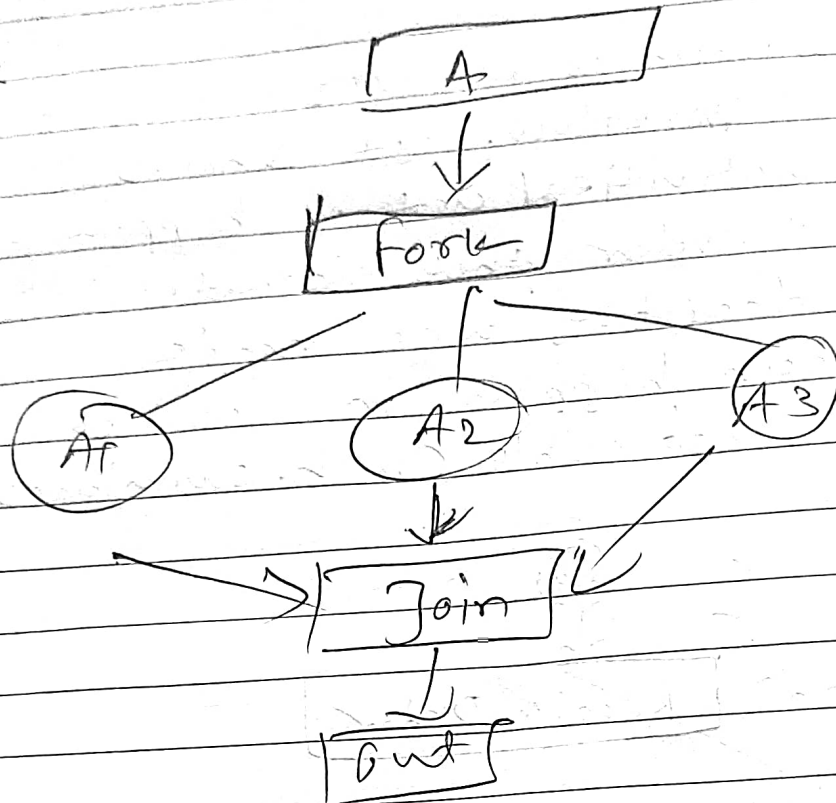
Flow final Node:



Merge Node

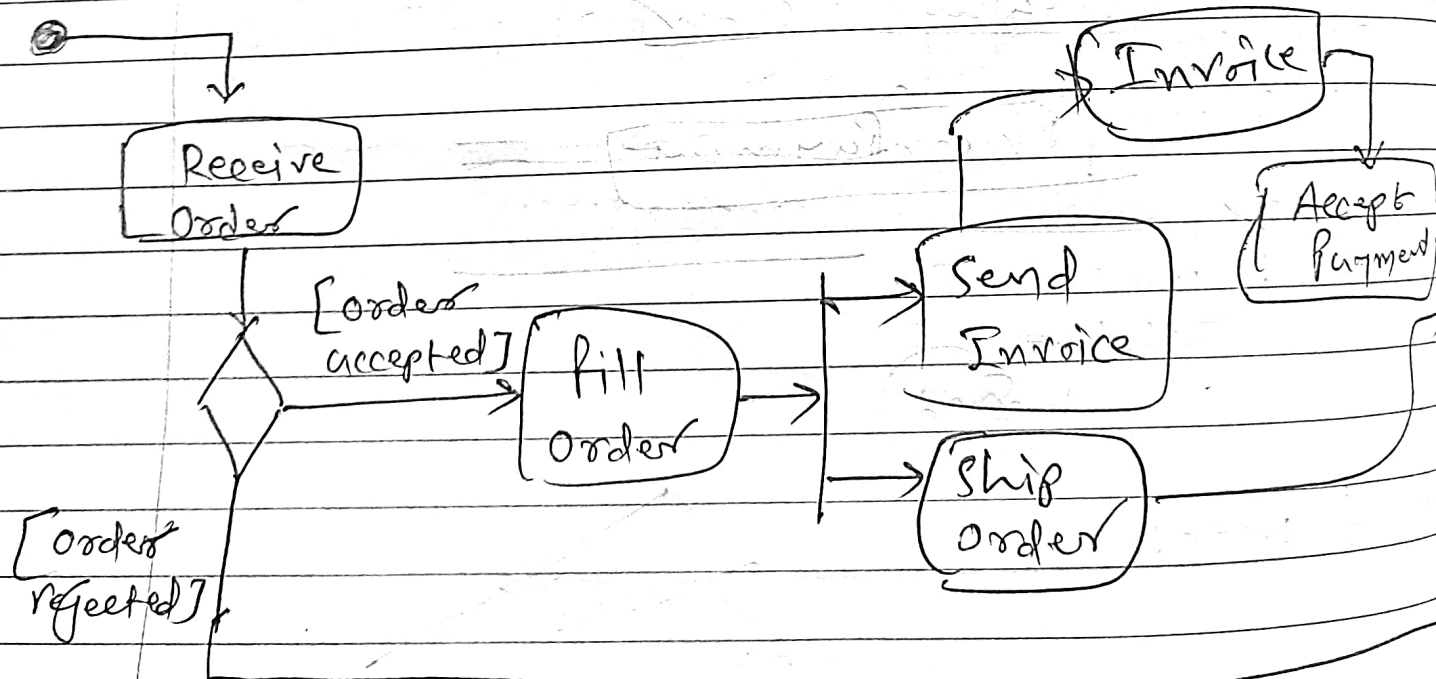


Fork Node



⇒

Process order

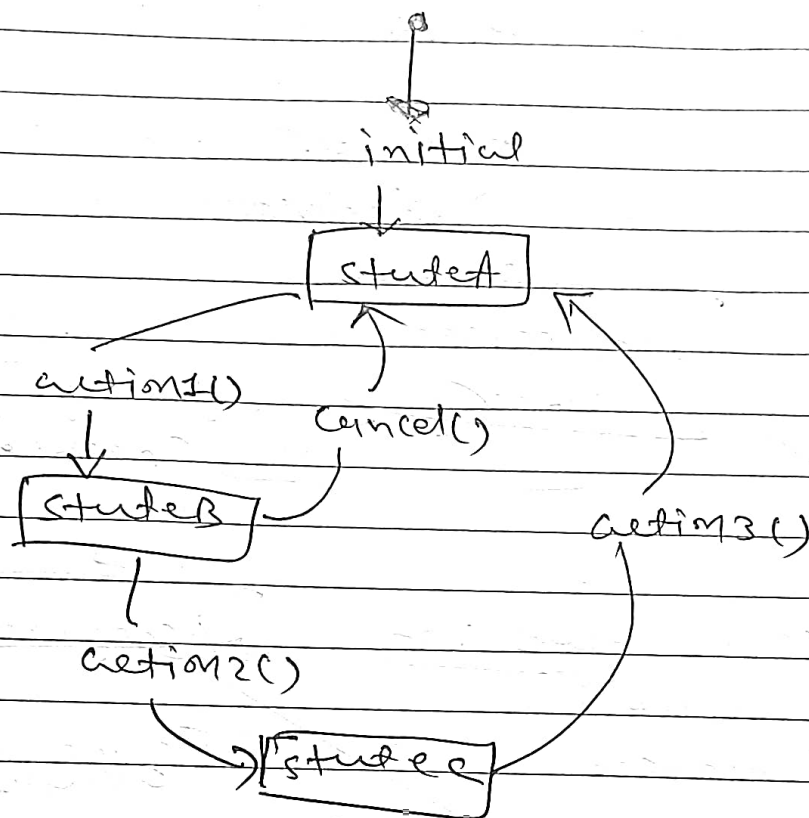


State Machine Diagram.

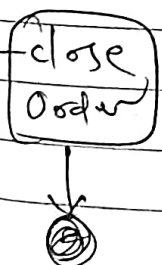
A state Diagram (also called a state Machine Diagram) is a UML Diagram that shows how an object or system changes its state over time in response to events.

It answers three key questions:

- What states can the system be in?
- What events cause transitions between states?
- What actions happen during those transitions?



- Benefits of state Diagrams
- Model object lifecycles clearly
- Catch missing transitions and ~~invalid~~ invalid states
- Bridge between requirements and implementation.



- Components of a State Machine Diagram

- state
- Initial state
- final state
- Transition
- Event
- Guard Condition
- Action

⇒ Online order system :-

